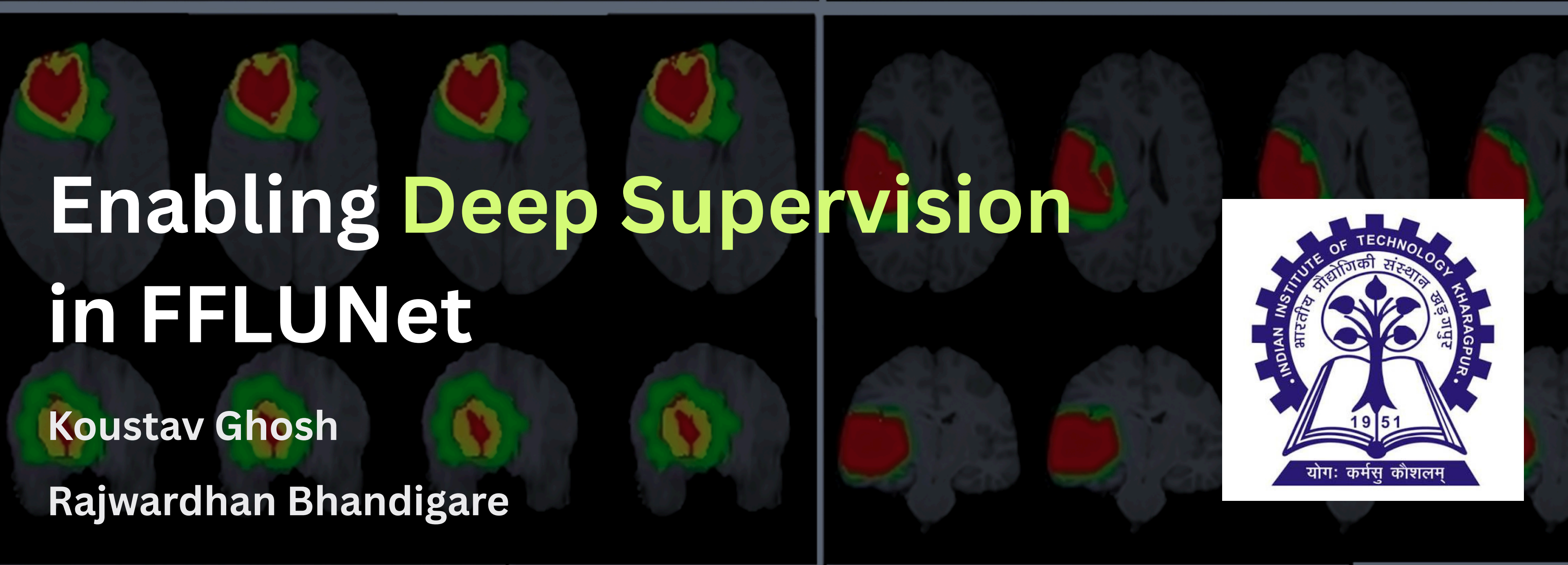
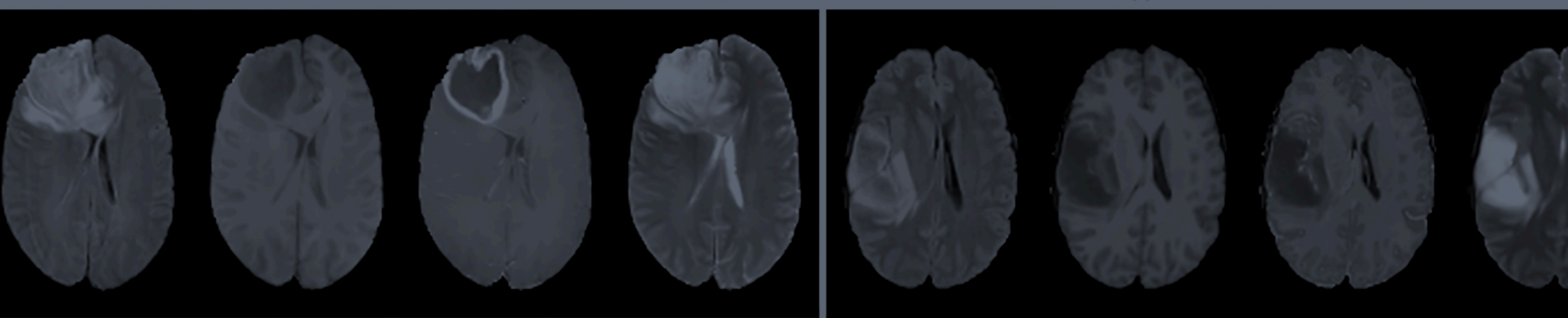
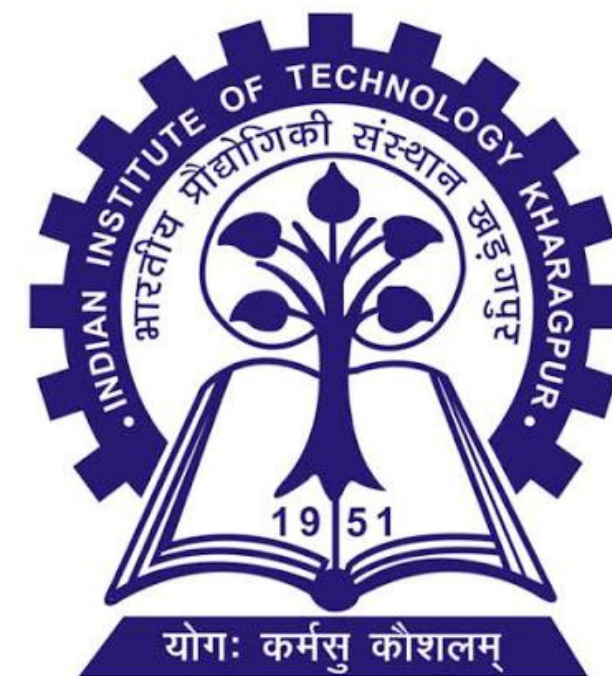




Enabling Deep Supervision in FFLUNet

Koustav Ghosh

Rajwardhan Bhandigare

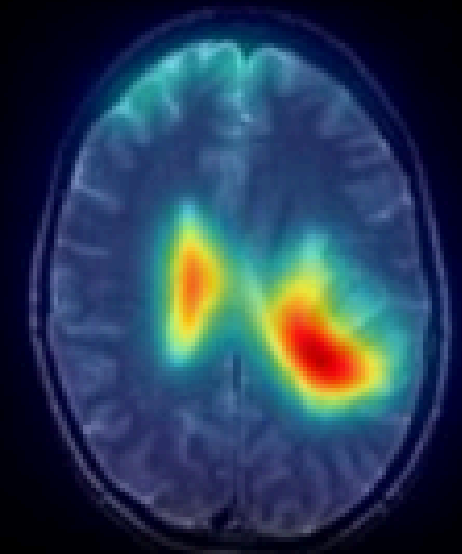
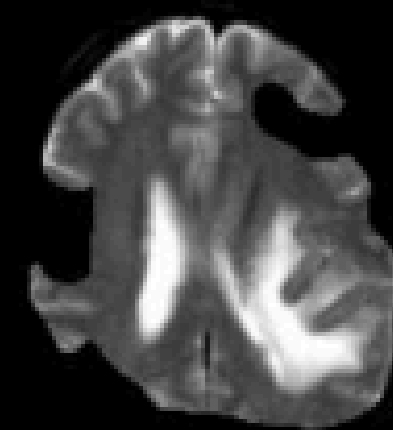
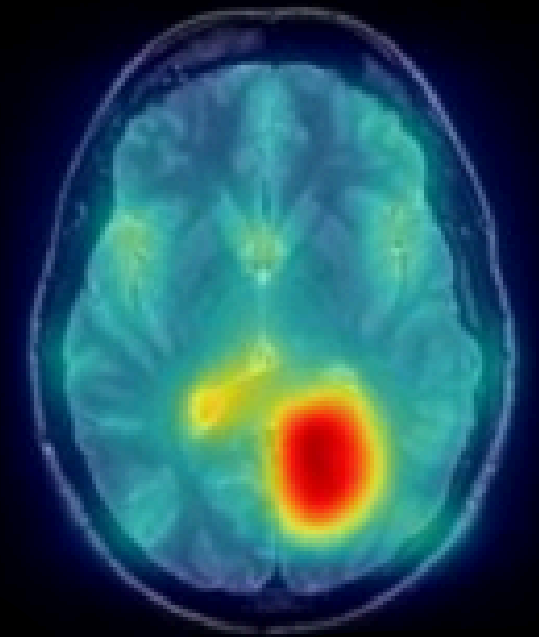
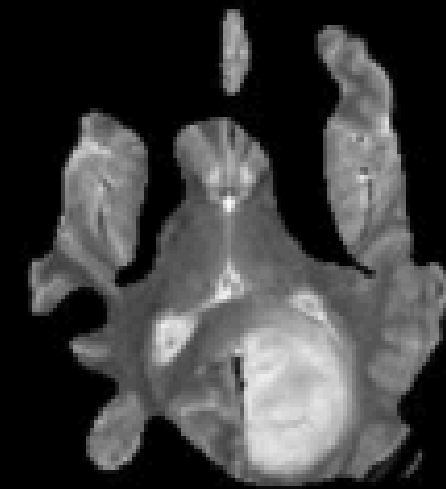


The Medical Context

Why Segmentation Matters in Precision Medicine

Medical Image Segmentation is the process of voxel-level classification for anatomical structures or pathologies from 3D scans (MRI/CT).

- **Critical for radiotherapy to define tumor boundaries.**
- **Critical for radiotherapy to define tumor boundaries.**
- **Reduces manual contouring time for doctors.**



Understanding the Dataset



<https://www.kaggle.com/datasets/awsaf49/brats20-dataset-training-validation>

Data Explorer

Version 1 (42.8 GB)

▼ BraTS2020_TrainingData

▼ MICCAI_BraTS2020_Tr

▶ BraTS20_Training_0

▶ BraTS20_Training_0

▶ BraTS20_Training_002 (5 files)

▶ BraTS20_Training_0

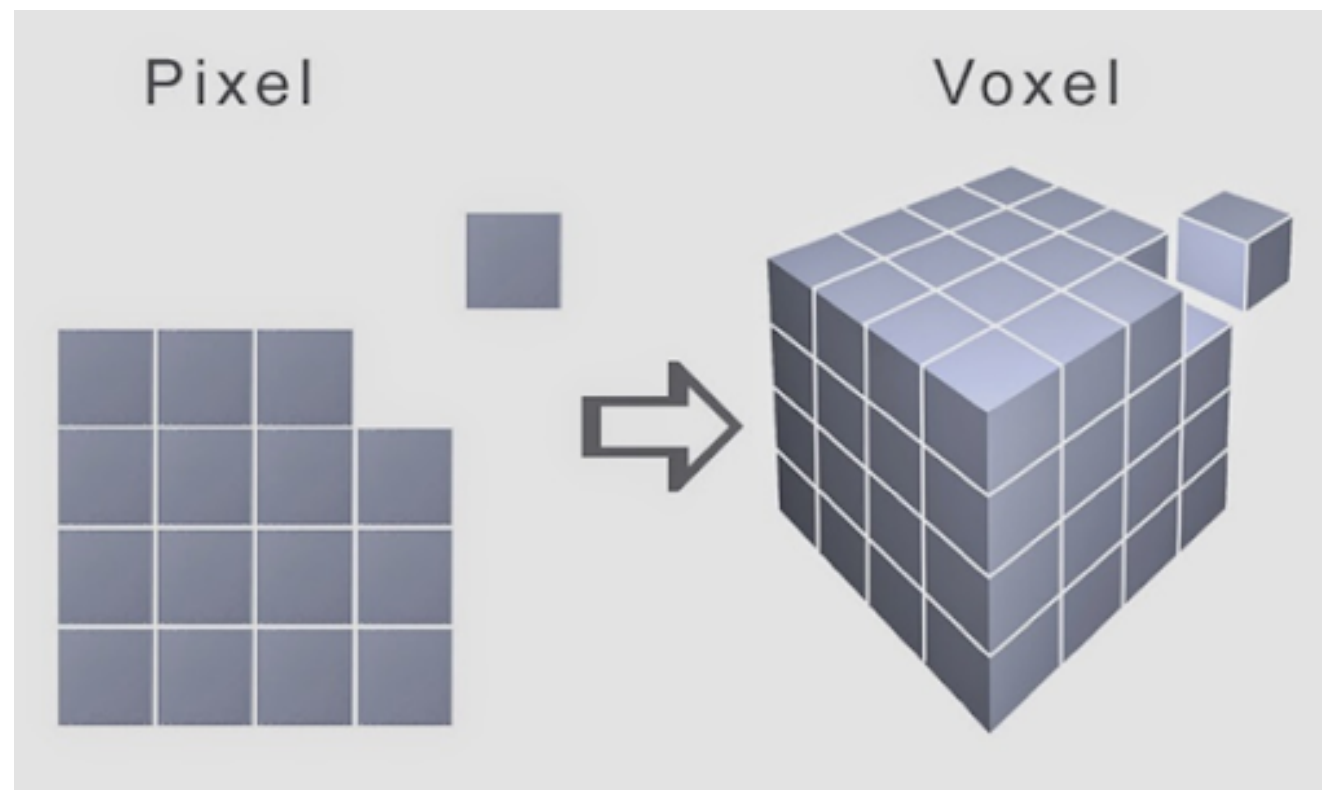
▶ BraTS20_Training_0

▶ BraTS20_Training_0

each folder corresponds to one entity
each folder has 5 files - 4 channels and 1 seg mask

- Multimodal scans available as NIfTI files (.nii.gz)
- Four 'channels' of information – 4 different volumes of the same region
 - i. Native (T1)
 - ii. Post-contrast T1-weighted (T1CE)
 - iii. T2-weighted (T2)
 - iv. T2 Fluid Attenuated Inversion Recovery (FLAIR) volumes

Understanding the Dataset



Files:

BraTS20_Training_001_flair.nii
BraTS20_Training_001_seg.nii
BraTS20_Training_001_t1.nii
BraTS20_Training_001_t1ce.nii
BraTS20_Training_001_t2.nii

Image Shapes:

T1 shape : (240, 240, 155)
T1CE shape : (240, 240, 155)
T2 shape : (240, 240, 155)
FLAIR shape : (240, 240, 155)
SEG shape : (240, 240, 155)

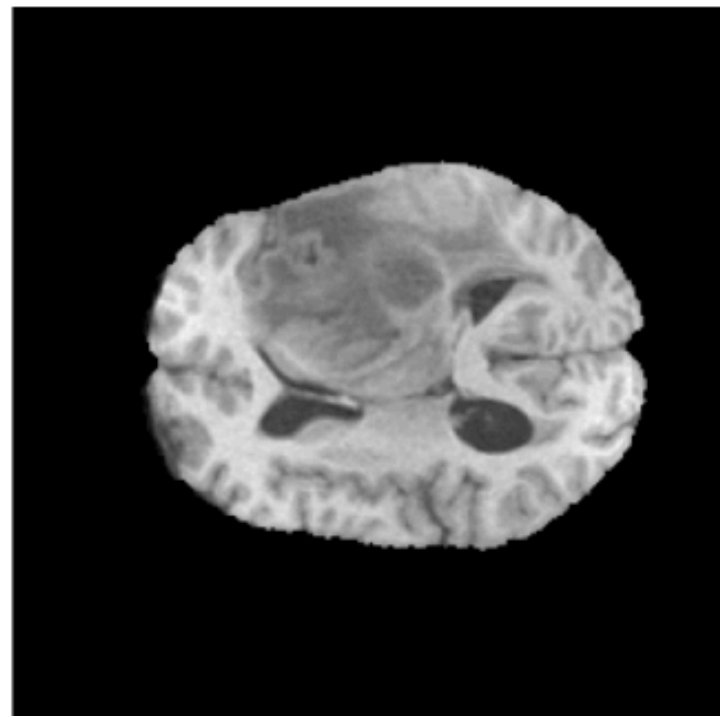
volume.shape = (240, 240, 155)

16 bit images

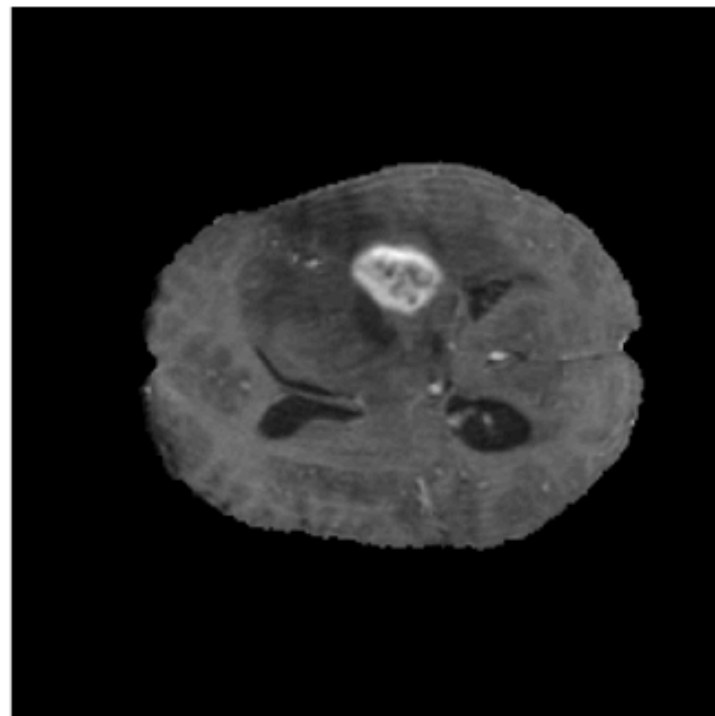
Note: Folder 355 has some weird name and so rename it or it gives error - no. of masks less than no. of images.

Visualizing slice: 77

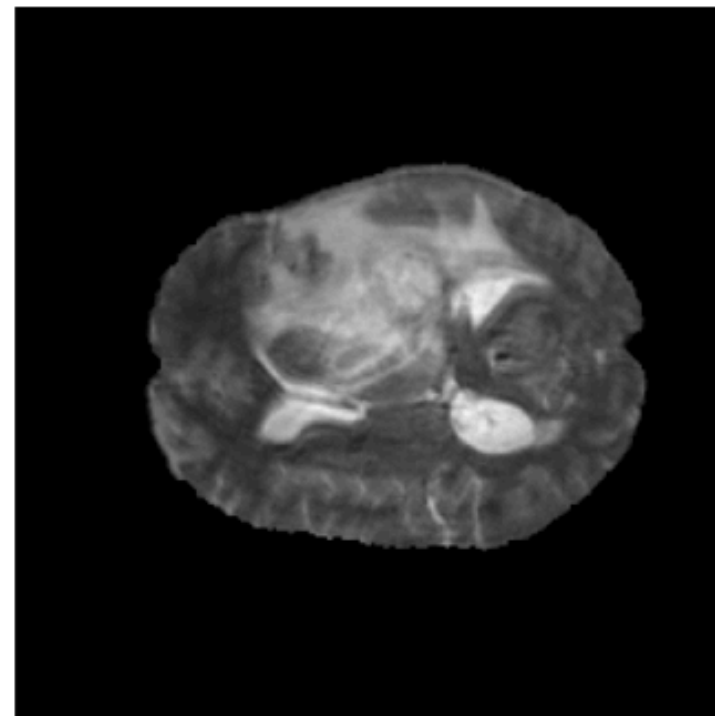
T1



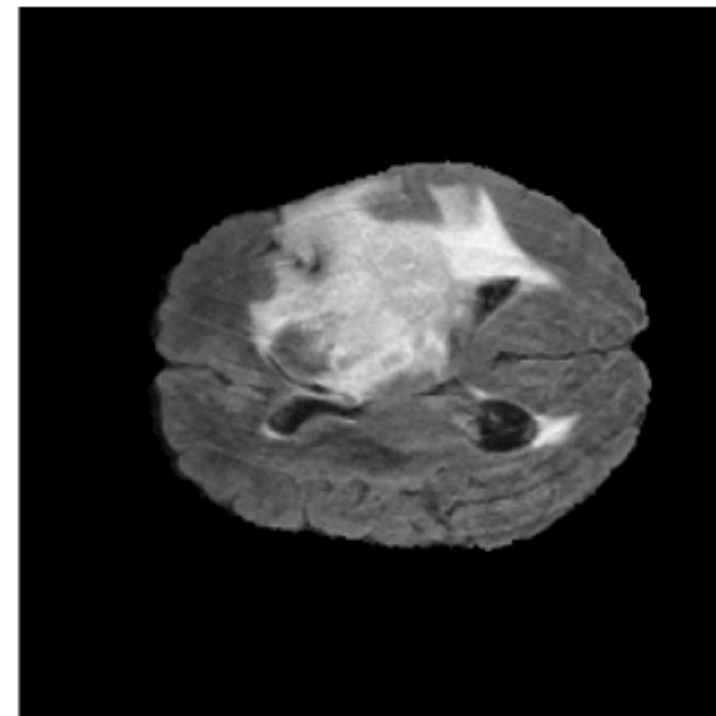
T1CE



T2



FLAIR



Segmentation



Data Preprocessing

Modality	Best for
T1	Brain anatomy
T1ce	Enhancing tumor
T2	Water content and edema
FLAIR	Edema and lesion visibility

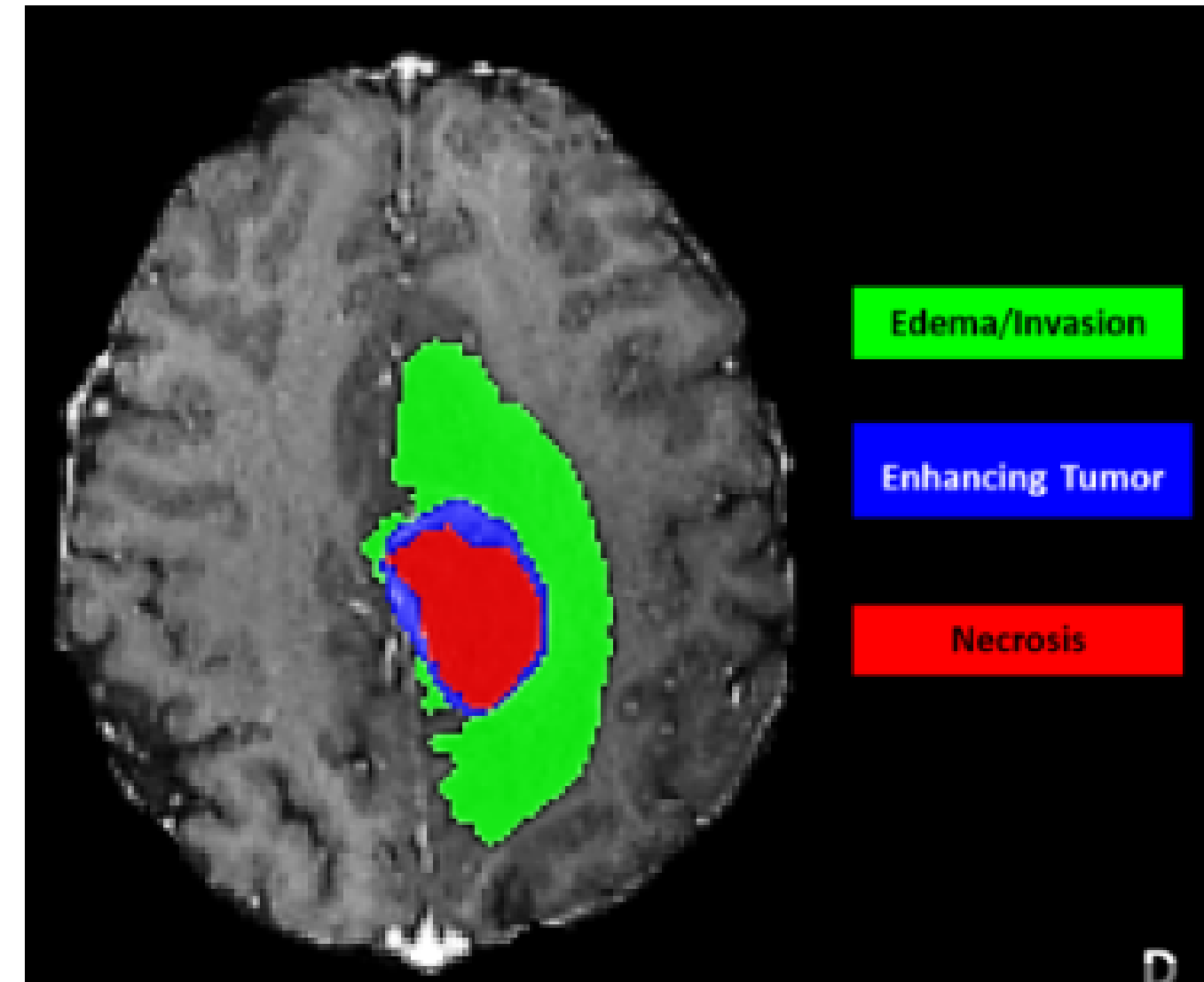
```
# label remap  
seg = seg.astype(np.uint8)  
seg[seg == 4] = 3
```

```
return image, seg
```

```
Image shape: (4, 240, 240, 155)
```

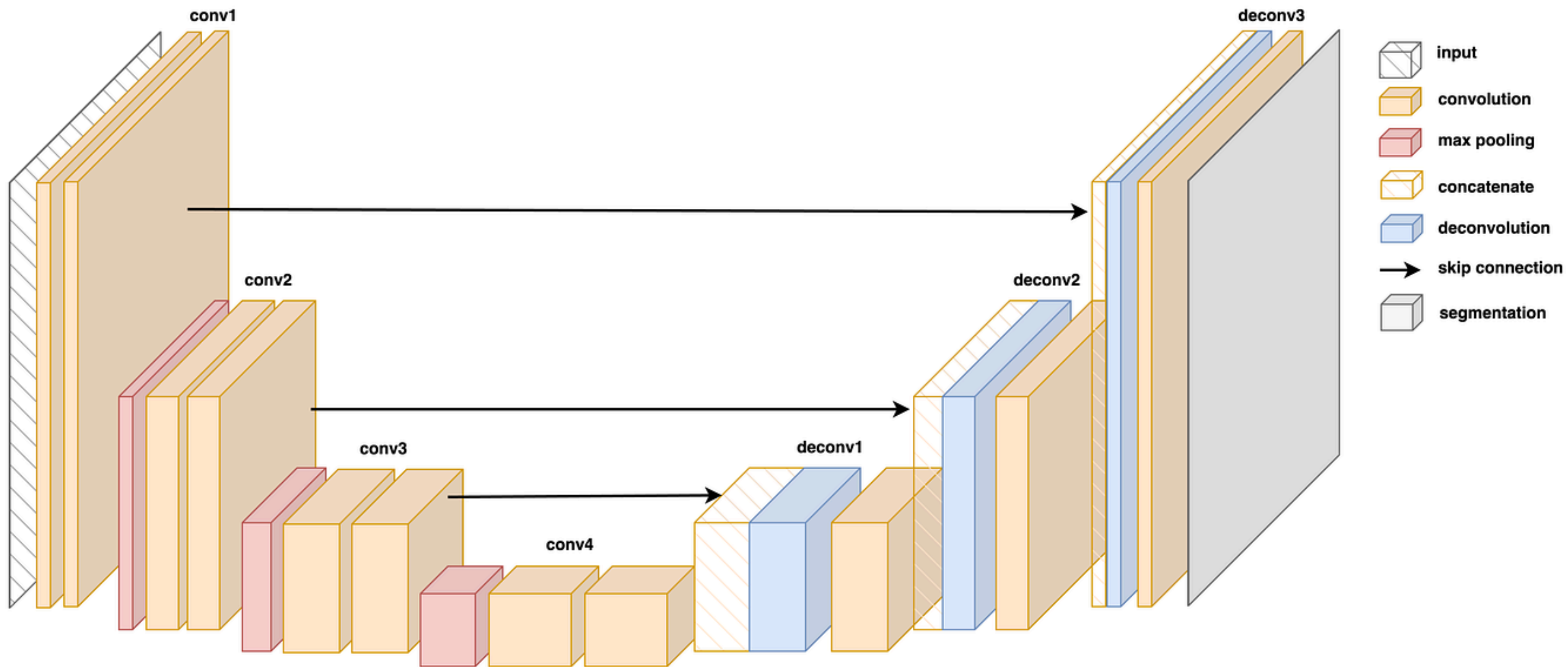
```
Mask shape : (240, 240, 155)
```

```
Mask labels: [0 1 2 3]
```



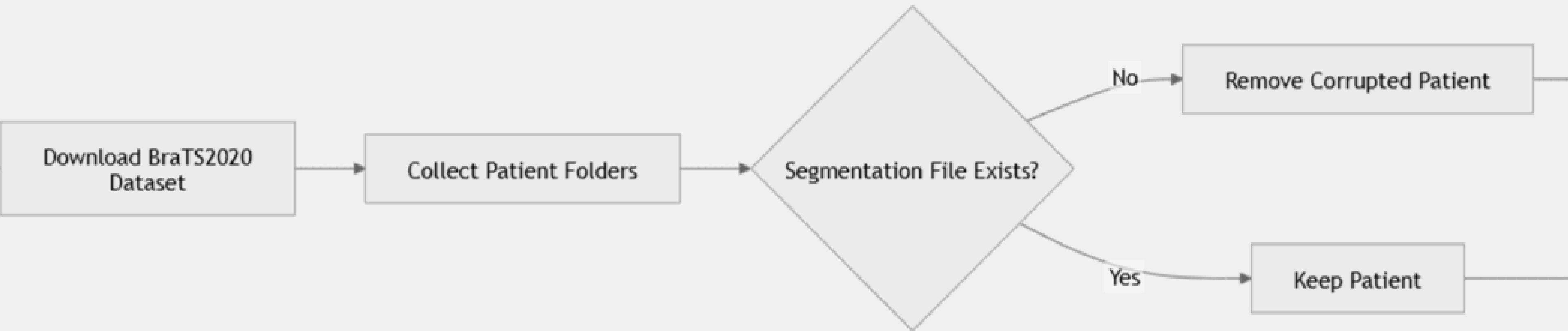
Reassign 4->3

UNet

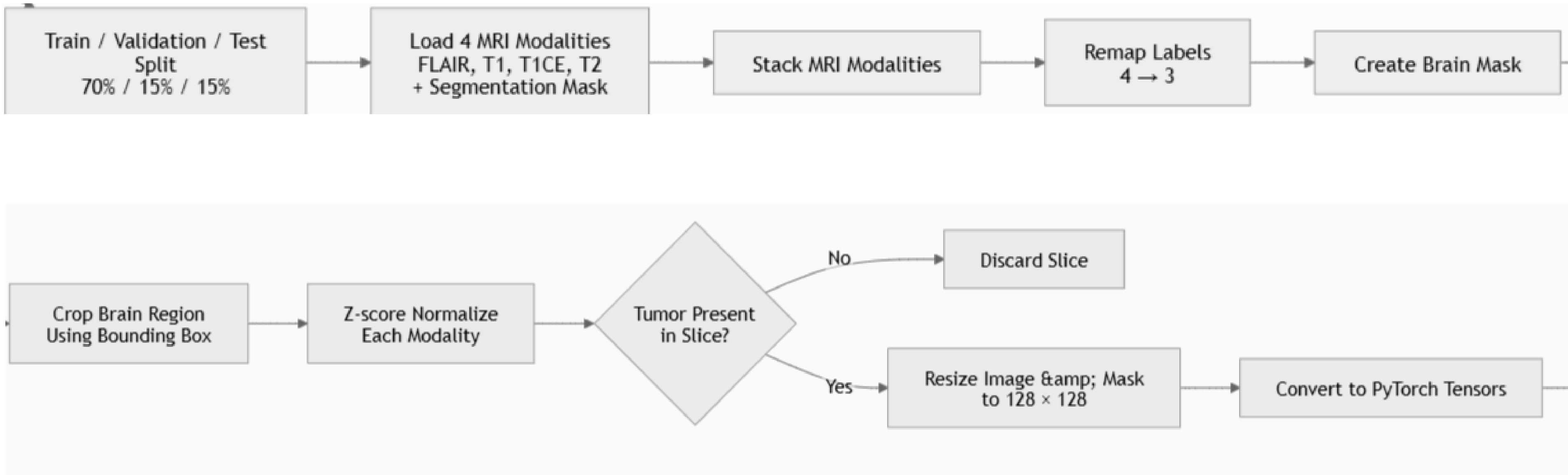


UNet

Created a 2D Unet from scratch and trained the layers on it



UNet



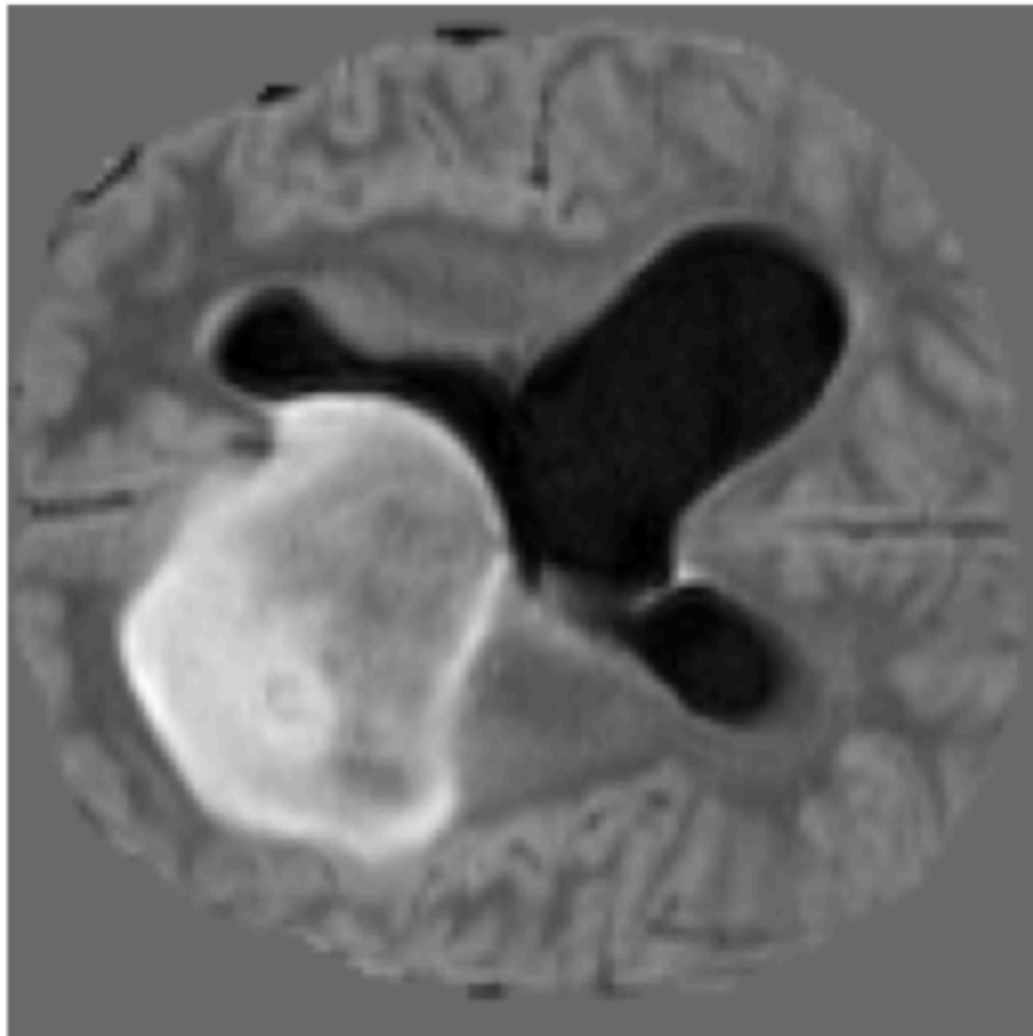
UNet

Patient 0: 68/146 tumor slices
Patient 1: 45/141 tumor slices
Patient 2: 80/139 tumor slices
Patient 3: 80/139 tumor slices
Patient 4: 43/141 tumor slices

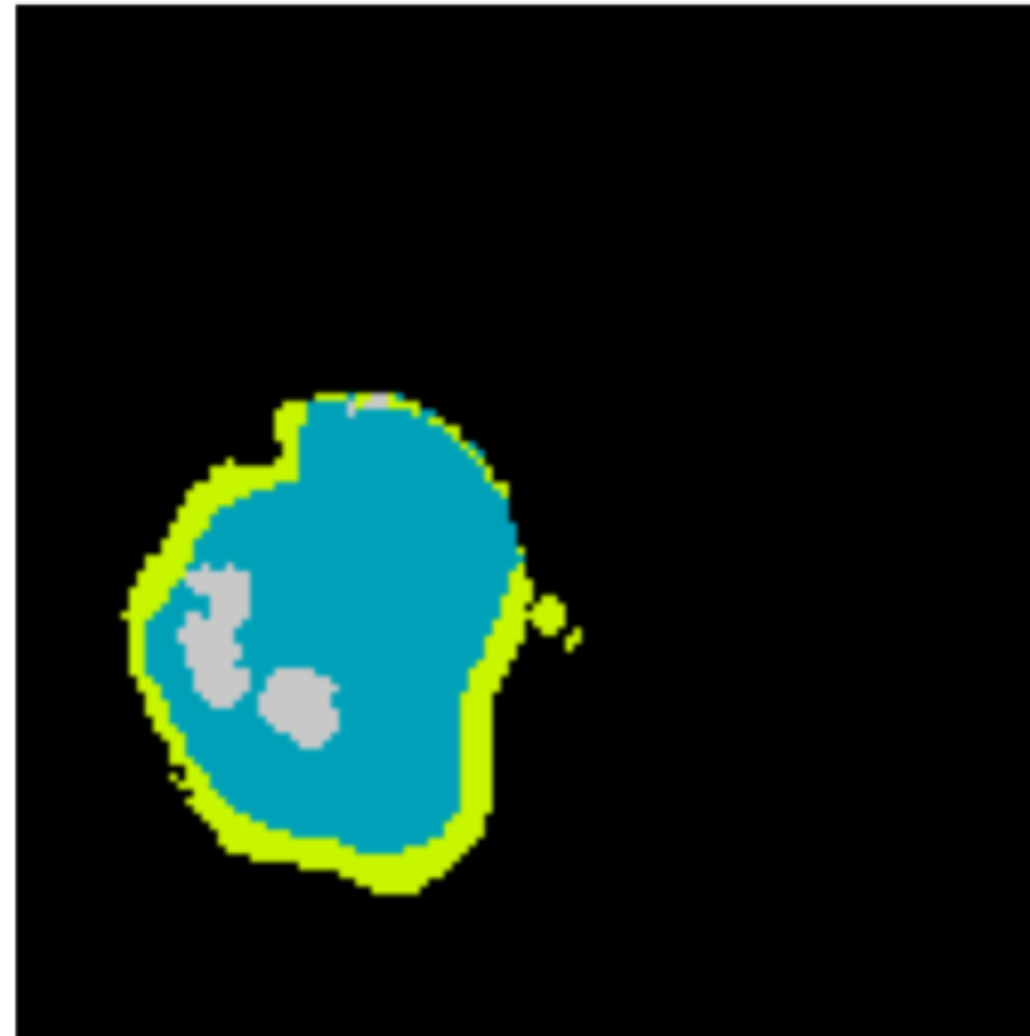
even after cropping there are
slices with no info, we only
take the slices with tumor seg
masks

Epoch 1/5		Loss: 0.1429
Epoch 2/5		Loss: 0.1092
Epoch 3/5		Loss: 0.0892
Epoch 4/5		Loss: 0.0734
Epoch 5/5		Loss: 0.0639

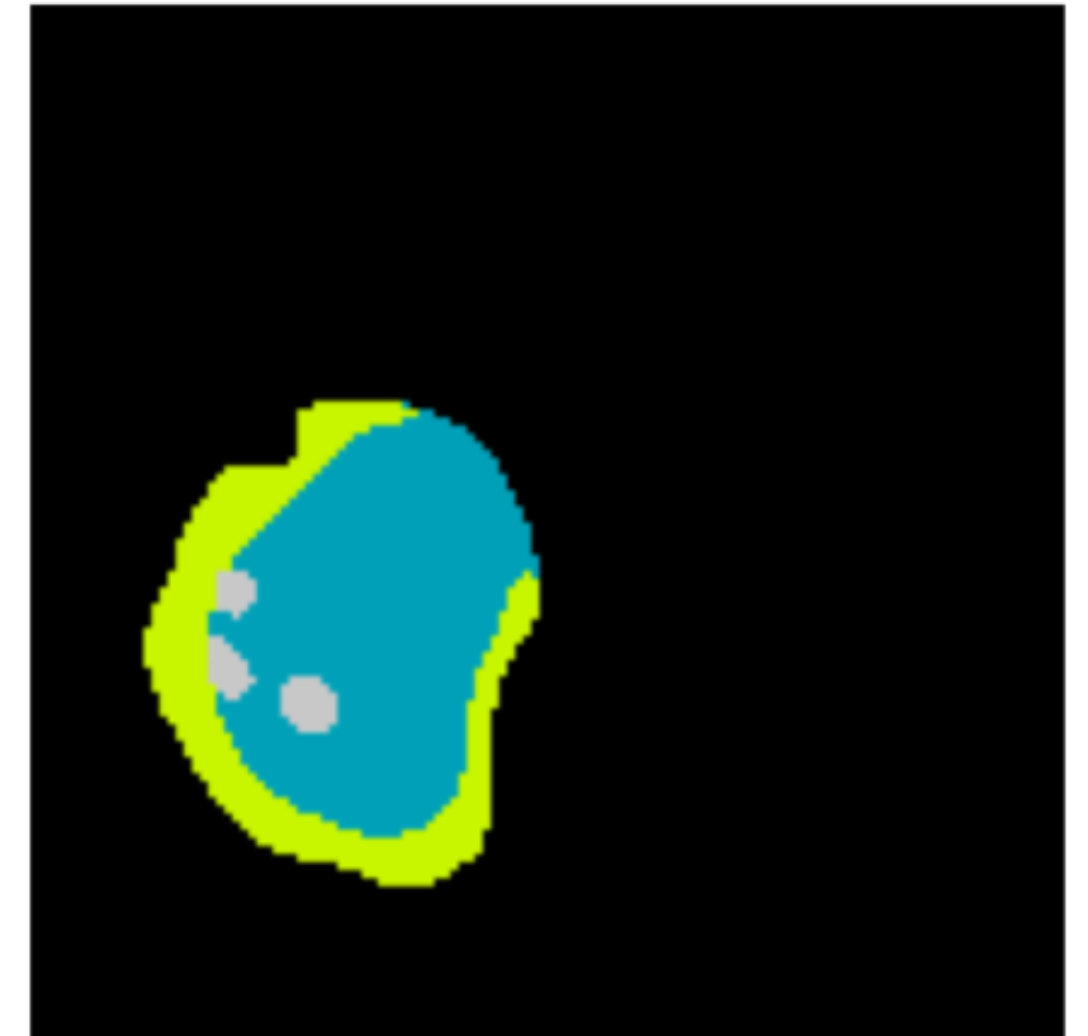
FLAIR



Ground Truth



Prediction



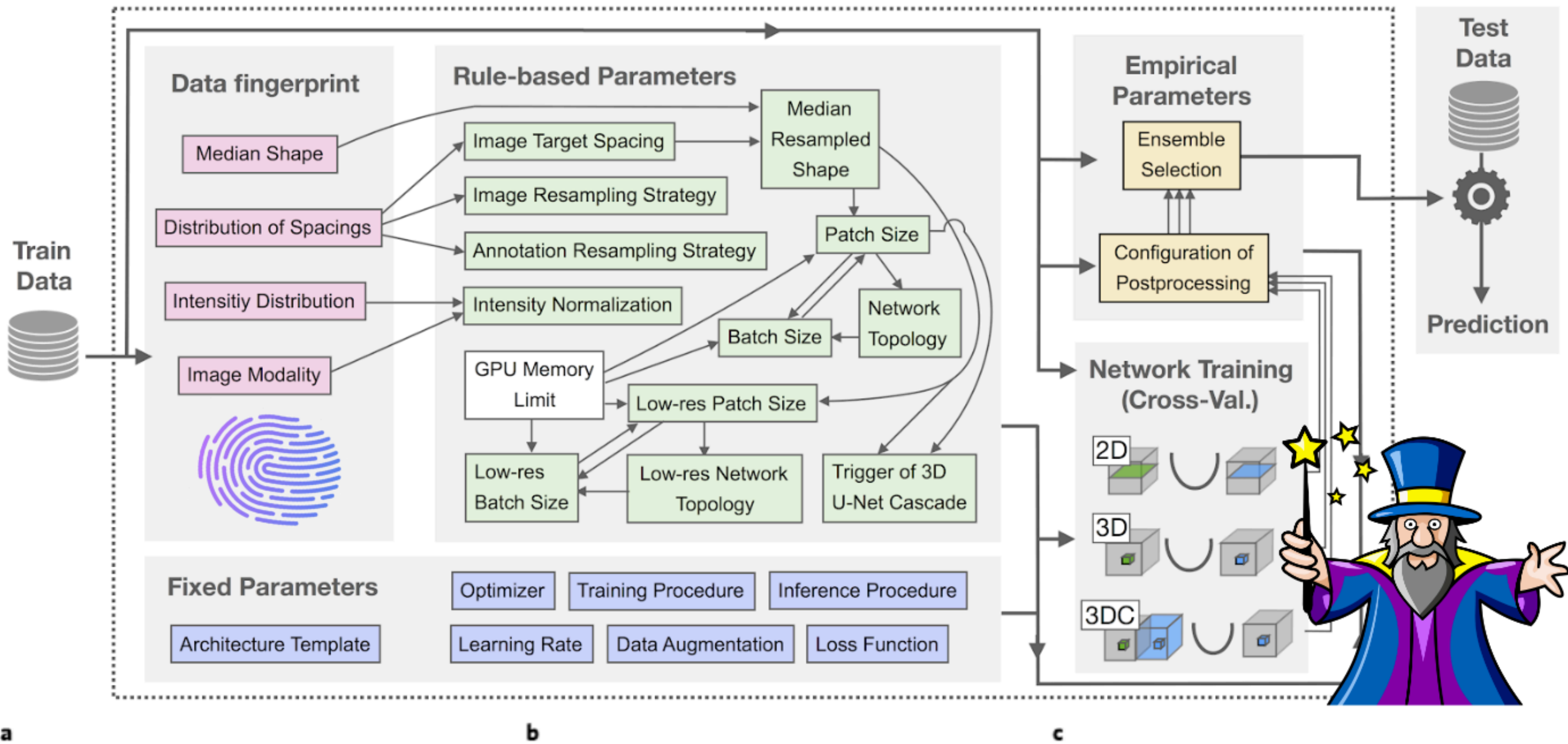
UNet

Step	2D U-Net	3D MONAI U-Net
Input	Tumor-containing axial slices	Entire 3D volume
Input size	(4, 128, 128)	(4, 128, 128, 128)
Model	Custom 2D U-Net	MONAI 3D U-Net
Loss	CrossEntropy + custom Dice	DiceCELoss
Batch size	8	1
Resize	2D slice-wise	Full 3D volume resize
Evaluation	Binary Dice	Dice, HD95, and Volumetric Similarity

3D UNet - MONAi

Epoch 1/5		Loss: 2.4472
Epoch 2/5		Loss: 2.3053
Epoch 3/5		Loss: 2.2367
Epoch 4/5		Loss: 2.1888
Epoch 5/5		Loss: 2.1512

nnUNet <https://github.com/mic-dkfz/nnunet>



NNUnet

Nnunet by default runs for 1000 epochs, so I didn't run it for that long, but just observed the initial few

It takes in a specific format ->

```
nnUNet_raw/  
├── Dataset001_BraTS2020/  
│   ├── imagesTr/  
│   ├── labelsTr/  
│   └── dataset.json
```

For each patient:

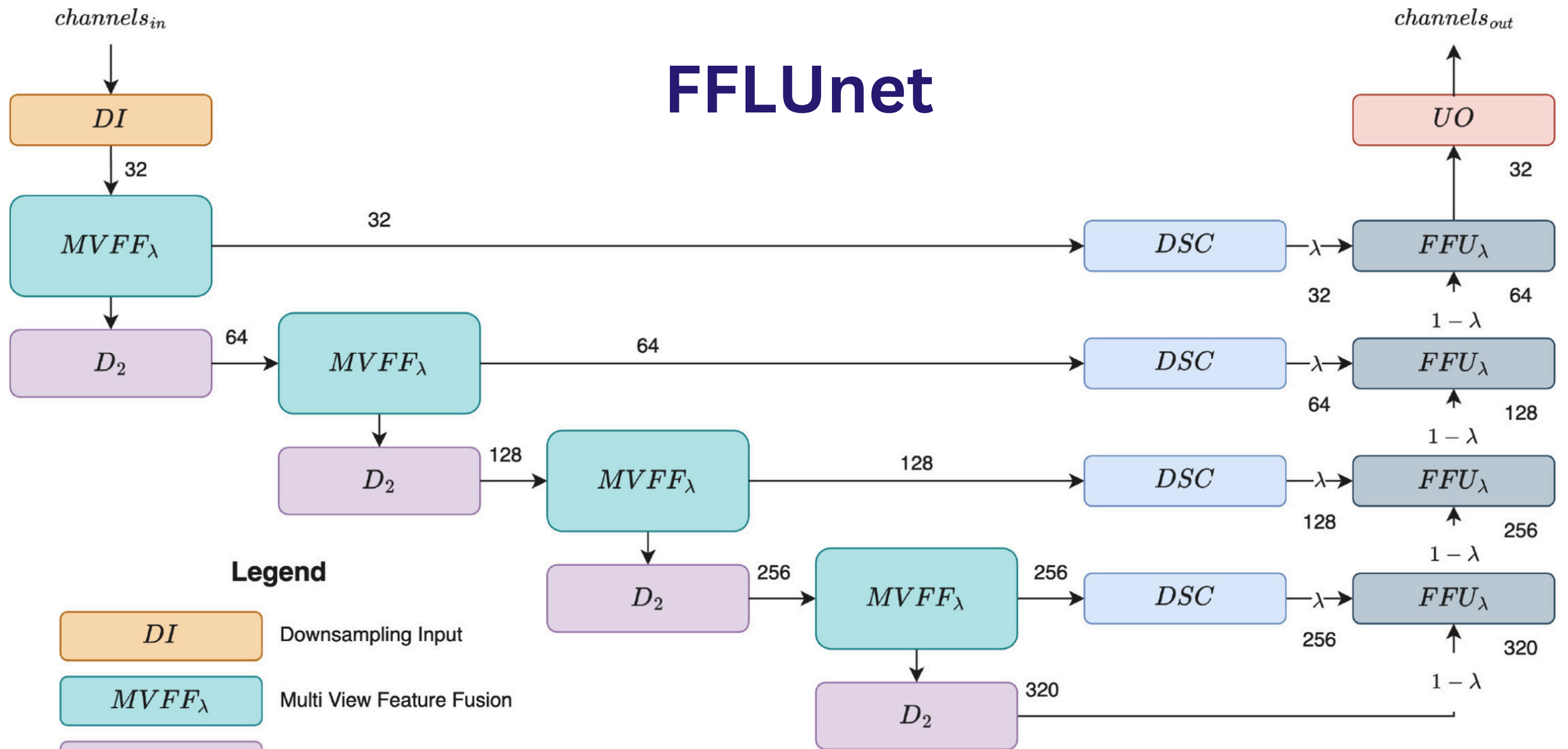
```
imagesTr/  
BraTS20_Training_001_0000.nii.gz ← FLAIR  
BraTS20_Training_001_0001.nii.gz ← T1  
BraTS20_Training_001_0002.nii.gz ← T1CE  
BraTS20_Training_001_0003.nii.gz ← T2  
  
labelsTr/  
BraTS20_Training_001.nii.gz ← segmentation
```

Epoch	Train Loss	Validation Loss	Dice Score 1	Dice Score 2	Dice Score 3	Avg. Dice (Approx.)
0	-0.222	-0.4716	0.4269	0.7401	0.8009	0.656
1	-0.4124	-0.4538	0.4511	0.6933	0.7979	0.6474
2	-0.4988	-0.54	0.5808	0.757	0.8121	0.6613
3	-0.5276	-0.5655	0.5456	0.7724	0.82	0.6664
4	-0.5552	-0.609	0.5377	0.8016	0.8416	0.6725
5	-0.5708	-0.6234	0.6858	0.814	0.8213	0.6826
6	-0.576	-0.6012	0.6142	0.8126	0.8194	0.6821

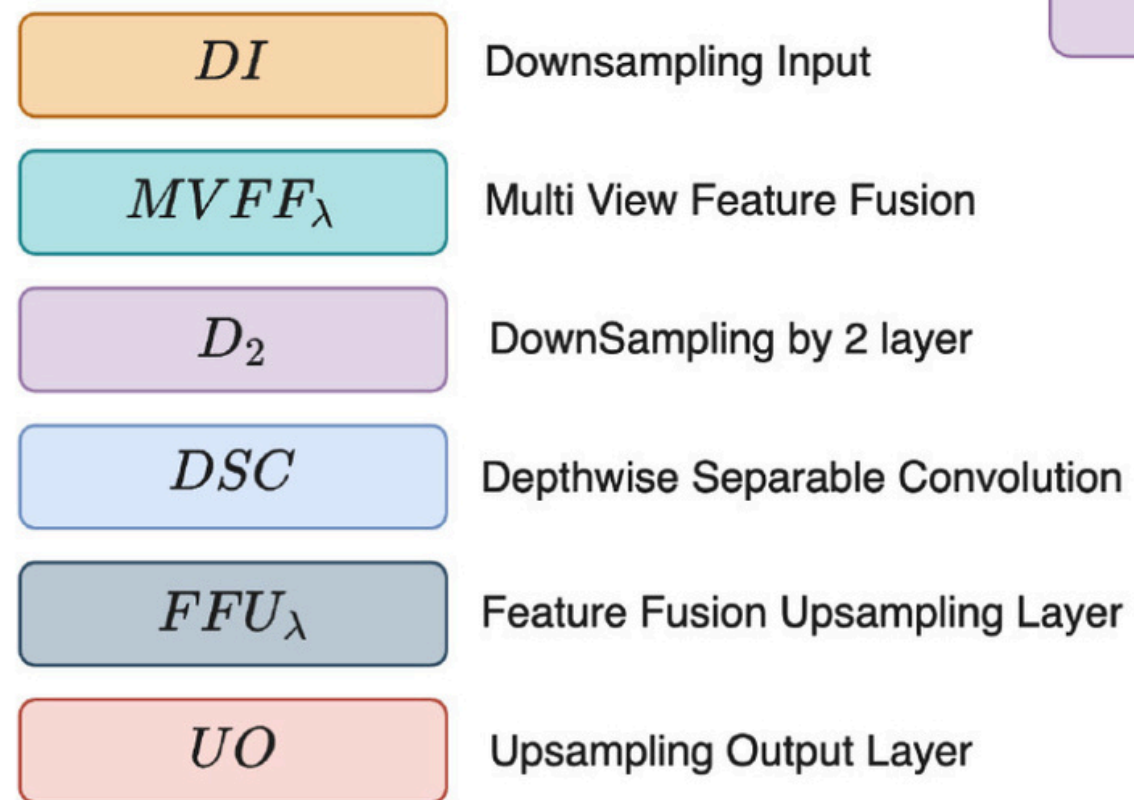
FFLUnet

FFLUNet is a novel, lightweight U-Net variant that incorporates multi-view feature fusion modules to enhance segmentation accuracy while reducing computational overhead. This model is designed for efficient and effective medical image segmentation and has been evaluated on multiple benchmark datasets.

FFLUNet



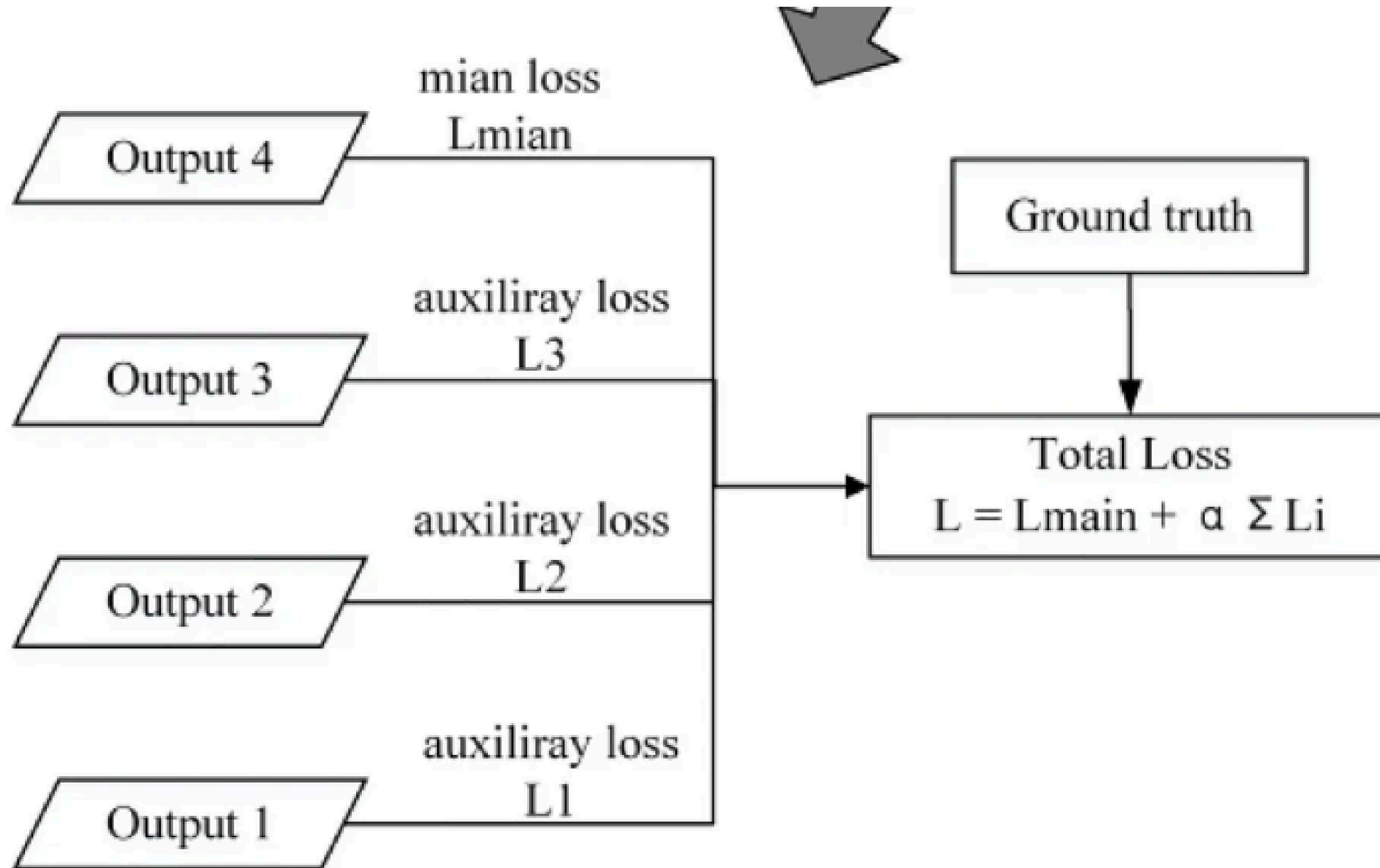
Legend



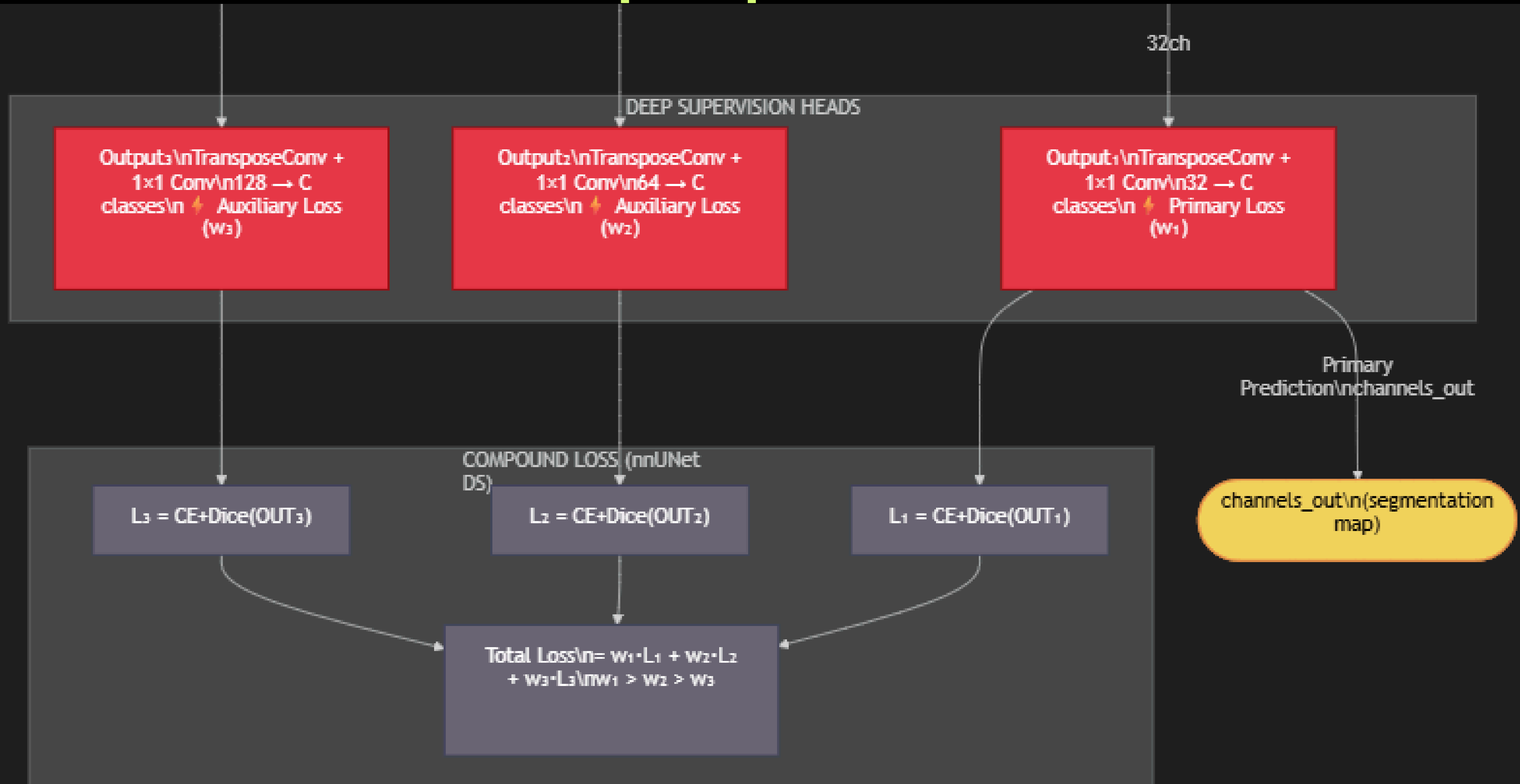
FFLUnet

Epoch	Learning Rate	Train Loss	Validation Loss	Dice Class 1	Dice Class 2	Dice Class 3	Mean Dice
0	0.01	0.2053	-0.068	0.2792	0.0103	0	0.0965
1	0.00999	-0.063	-0.1757	0	0.5587	0.4256	0.1197
2	0.00998	-0.1056	-0.2121	0.0174	0.6159	0.5189	0.1461
3	0.00997	-0.1947	-0.3886	0.3289	0.7185	0.7427	0.1912
4	0.00996	-0.2828	-0.4493	0.339	0.7018	0.7924	0.2331

Deep Supervision



Deep Supervision



Architecture Comparison

Aspect	nnUNet	FFLUNet (no DS)	FFLUNet (with DS)
Skip connection	Concat	Weighted add (λ)	Weighted add (λ)
Encoder block	Standard conv	MVFF (local+global)	MVFF (local+global)
Decoder block	Standard conv	DSC (lightweight)	DSC (lightweight)
Output heads	1 (+ DS heads if enabled)	1	3 (multi-scale)
Training loss	Compound (DS on by default)	Single	Compound (weighted)
Gradient to encoder	Through decoder path	Through decoder path	Through decoder + auxiliary heads
Trainer class	<code>nnUNetTrainer</code>	<code>nnUNetTrainerNoDeepSupervision</code>	<code>nnUNetTrainer</code>

Deep Supervision

Epoch	Learning Rate	Train Loss	Validation Loss	Dice Class 1	Dice Class 2	Dice Class 3	Mean Dice
0	0.01	0.2635	-0.0328	0.2263	0.4317	0	0.2193
1	0.00999	-0.0633	-0.0457	0	0.4506	0.2947	0.2222
2	0.00998	-0.093	-0.1661	0	0.4814	0.5382	0.234
3	0.00997	-0.1398	-0.2162	0.1415	0.6245	0.66	0.2581
4	0.00996	-0.1688	-0.2466	0.3915	0.5298	0.7914	0.2894
5	0.00995	-0.2139	-0.2675	0.3776	0.5687	0.8101	0.319
6	0.00995	-0.2382	-0.3278	0.3737	0.657	0.7701	0.3471
7	0.00994	-0.2549	-0.313	0.5491	0.627	0.6874	0.3745
8	0.00993	-0.3043	-0.3558	0.568	0.6786	0.79	0.405
9	0.00992	-0.3193	-0.4489	0.563	0.7512	0.8065	0.4352
10	0.00991	-0.347	-0.3873	0.4739	0.6536	0.783	0.4553

Comparison

Metric	FFLUNET	FFLUNET + DS
Training Loss	-0.1688	-0.2828
Validation Loss	-0.2466	-0.4493
Dice (Class 1)	0.3915	0.339
Dice (Class 2)	0.5298	0.7018
Dice (Class 3)	0.7914	0.7924
Mean Dice (EMA)	0.2894	0.2331

Model A: higher Mean Dice

"labels": {"background": 0, "necrotic": 1, "edema": 2, "enhancing": 3}

Model B: better loss optimization and stronger Class 2 performance, but lower overall Dice

Conclusion

We trained two versions of FFLUNet on brain tumor MRI scans to segment three tumor regions: the necrotic core (dead tissue), edema (swelling), and enhancing tumor (active cancer cells).

The standard FFLUNet learns from only the final output. The deep supervision version gets extra feedback signals from intermediate layers during training, forcing the network to build useful features at every scale, not just the last one.

When we compared them, the deep supervision model fit the training data better and also generalised better to unseen scans — both its training and validation losses were lower. On actual segmentation quality, it was dramatically better at detecting edema (the large swelling region), essentially the same on the enhancing tumor, and slightly worse on the small necrotic core — which makes sense because deep supervision helps most on large structures visible at multiple scales, but can dilute gradient signal for tiny regions.

The one metric that looked worse for the DS model was the Mean Dice EMA, but that's misleading — it's a rolling average computed across all training epochs, and the DS model starts slower because it's juggling three loss signals at once from the beginning. It looks worse mid-training but converges to a better place.

Bottom line: Adding deep supervision made FFLUNet a meaningfully better model overall, especially for the largest and most diffuse tumor region. The trade-off is a slightly harder training process and marginally weaker detection of the smallest tumor subregion.

References

[**https://github.com/mic-dkfz/nnunet**](https://github.com/mic-dkfz/nnunet)

[**https://github.com/Dutta-SD/FFLUNet**](https://github.com/Dutta-SD/FFLUNet)

[**https://drive.google.com/drive/u/0/folders/1rCQj6cNHuThNUKVWf0dTqete39zo5WJs**](https://drive.google.com/drive/u/0/folders/1rCQj6cNHuThNUKVWf0dTqete39zo5WJs)

[**https://github.com/RAJWARDHAN-B/FFLUNet**](https://github.com/RAJWARDHAN-B/FFLUNet)

branch: add-deep-supervision